👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.AT.8

# Write and Solve Equations Using Addition or Subtraction

Lesson 8-2 · Teacher Copy

**Name:** _____ **Date:** _____ **Class:** _____

## My Notes

**Level 3 Enrichment**

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can solve one-step addition and subtraction equations using inverse operations.**

**Language Objective: I can explain my steps using the words equation, inverse operation, isolate, and solution.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

**Equation** · **Inverse operation** · **Isolate** · **Solution** · **One-step equation** · **Balance** · **Subtraction Property of Equality** · **Addition Property of Equality**

*Level 3: try the blanks from memory first, then check the bank.*

**1** A math sentence with an equal sign showing two equal amounts is an _____.

2. An operation that undoes another, like subtraction undoing addition, is an _____.

3. To get the variable alone on one side of the equation is to _____ it.

4. The value of the variable that makes the equation true is the _____.

5. An equation solved in a single step is a _____.

6. Doing the same thing to both sides to keep them equal keeps the equation in _____.

7. Subtracting the same amount from both sides uses the _____.

8. Adding the same amount to both sides uses the _____.

## Write About the Math

**How far has the hiker walked? How did you use inverse operations?**
*¿Qué distancia ha caminado la excursionista? ¿Cómo usaste las operaciones inversas?*

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Equation** *Ecuación*
- ☐ **Inverse operation** *Operación inversa*
- ☐ **Isolate** *Despejar*
- ☐ **Solution** *Solución*
- ☐ **One-step equation** *Ecuación de un paso*

**Model:** *I can undo the addition to get the variable by itself.* — Students recognize 23 is added to n and that subtracting 23 (the inverse operation) from both sides will isolate n.

## Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The hiker walked ___ miles because d + 4.5 = 12 means d = 12 − ___ = ___.**
- **My answer is ___ because ___.**
  *Mi respuesta es ___ porque ___.*

## Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ___.**
  *Mi afirmación es ___.*
- **My evidence is ___, so ___.**
  *Mi evidencia es ___, entonces ___.*

**Explain**

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

## Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Inverse operation
3. **Notes 3:** Isolate
4. **Notes 4:** Solution
5. **Notes 5:** One-step equation
6. **Notes 6:** Balance
7. **Notes 7:** Subtraction Property of Equality

8. **Notes 8:** Addition Property of Equality
9. **Watch for this mistake:** *A common mistake is applying the same operation instead of the inverse — for example, solving x − 7 = 15 by subtracting 7 from both sides (getting the wrong x = 8) instead of adding 7 to both sides (the correct x = 22). Another common mistake is doing the inverse operation to only one side, like solving x + 6 = 14 by subtracting 6 from the left but forgetting the right side. Before you submit, name the operation shown in the equation, choose its inverse, apply it to BOTH sides, and check by substituting your answer back into the original equation.*
10. **Try It 1:** C. n = 13 — *Add 4 to both sides: n = 9 + 4 = 13. Check: 13 − 4 = 9 ✓*
11. **Try It 2:** B. 34 — *Subtract 16 from both sides: t = 50 − 16 = 34. Check: 34 + 16 = 50 ✓*